



# Statistics of Financial Markets

## Chapter 10: VaR, ES and backtesting



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ BCBS: Basel Committee on Banking Supervision
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIS: Bank for International Settlements
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CLT: Central Limit Theorem
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ HS: Historical Simulation
- ▣ LR: Likelihood Ratio (test)
- ▣ POF: Proportion Of Failures (Kupiec test)
- ▣ POT: Peaks Over Threshold
- ▣ SFE: Statistics of Financial Markets, Exercises (Quantlet collection of the textbook)
- ▣ US: United States

## Today's Question and Route

- **Question:** how much can a portfolio lose tomorrow, and how do we know that the answer was right?
  - ▶ Chapter 5 described the tails, Chapter 9 the volatility of tomorrow; this chapter turns both into risk numbers and tests them
- **Route** of the chapter
  - ▶ VaR (value at risk) and ES (expected shortfall): definitions, coherence, elicibility
  - ▶ estimation: historical simulation, Normal, Student-t, Cornish–Fisher, EVT, GARCH and filtered historical simulation
  - ▶ portfolios: variance–covariance, tail dependence, a first look at copulas
  - ▶ backtesting: Kupiec, Christoffersen, the Basel traffic light, ES backtesting

## Learning Outcomes

- ▣ Define VaR and ES at a tail probability  $\alpha$  (VaR 1%, ES 2.5%) and compute them for the Normal and Student-t distributions
- ▣ State the axioms of a coherent risk measure and show with a counterexample that VaR is not subadditive
- ▣ Estimate VaR and ES by historical simulation, parametric methods, Cornish–Fisher, EVT, GARCH and filtered historical simulation
- ▣ Compute a portfolio VaR and explain why correlation does not measure the risk of joint crashes
- ▣ Backtest a VaR model with the Kupiec and Christoffersen tests and the Basel traffic light

## Reading and Tools

- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 16 (VaR and backtesting), Ch. 17 (copulae and VaR), Sections 18.1 and 18.4 (extreme risks)
  - ▶ exercises with solutions: [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 16
  - ▶ risk measures and copulas: [McNeil, Frey & Embrechts \(2015\)](#), Ch. 2 and 7
- Python Quantlets of this chapter: [Quantlets/Ch\\_10](#)
  - ▶ ported from the SFE Quantlets [VaRest](#), [SFEVaRbank](#), [SFEVaRtimeplot](#), [SFEVaRqqplot](#) and [SFEvar\\_pot\\_backtesting](#)
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Measuring Statistical Risk](#)

## The 4:15 Report and RiskMetrics

- Around 1990: J.P. Morgan's chairman, Dennis Weatherstone, asks for one number, every day at 4:15 pm, summarising how much the bank could lose the next day
  - ▶ the answer became the **value at risk** (VaR): a loss that is exceeded only with a small probability
- October 1994: the methods and data are published free of charge as RiskMetrics [J.P. Morgan/Reuters \(1996\)](#)
  - ▶ Normal returns, EWMA volatility ([Chapter 8](#)), VaR at a 5% tail probability over one day
  - ▶ VaR became the common language of banks, regulators and fund managers



The J.P. Morgan building at 23 Wall Street, New York

Photo: [Beyond My Ken \(2012\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

## From RiskMetrics to Basel

- The **BCBS** (Basel Committee on Banking Supervision) sits at the **BIS** (Bank for International Settlements) in Basel [history of the Basel Committee](#)
  - ▶ 1996: banks may compute market-risk capital with their own VaR models [Basel Committee \(1996a\)](#)
  - ▶ rule: VaR 1% over 10 days, times a multiplier of at least 3; the multiplier grows when the model fails the backtest [Basel Committee \(1996b\)](#)
- 2019: the **FRTB** (Fundamental Review of the Trading Book) [Basel Committee \(2019\)](#)
  - ▶ capital from ES 2.5% instead of VaR 1% [Basel Framework, MAR33](#); backtesting still counts VaR exceptions [Basel Framework, MAR32](#)



The BIS tower in Basel, seat of the Basel Committee

Photo: [Taxiarchos228 \(2014\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

## Two Storms: 2008 and 2020



Lehman Brothers headquarters, New York, 15 September 2008

Photo: Robert Scoble (2008); CC BY 2.0; Wikimedia Commons



The Financial District of New York, 25 March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

- The worst day of the S&P 500 since 2000:  $-12.8\%$  on 16 March 2020; a Normal model with the sample volatility gives such a day a probability of practically zero
- A risk measure must be judged in such periods: we use 2008 and 2020 as stress tests throughout the chapter

## Recap: Measuring Tail Risk

- VaR was born as a daily summary of the next day's loss (RiskMetrics, 1994)
- Basel: VaR 1% for capital (1996), ES 2.5% for capital and VaR 1% for backtesting (FRTB, 2019)
- Crises are the real test of a risk model

## Losses, Horizon and Level

- $X$ : the **P&L** (profit and loss) or the return of a position over a horizon  $h$  (here one day, in %)
  - ▶ the loss is  $L = -X$ : a positive number when the position loses money
- The **level**  $\alpha$  is the probability of the tail:  $\alpha = 1\%$  or  $\alpha = 2.5\%$ 
  - ▶ we write **VaR 1%** and **ES 2.5%**: the bad outcomes have probability 1% and 2.5%
  - ▶ the number in the name is always the tail probability: VaR 1% is exceeded with probability 1%
- $q_\alpha(X) = \inf\{x : P(X \leq x) > \alpha\}$ : the  $\alpha$ -**quantile** of  $X$  ([Chapter 2](#))
  - ▶ for a continuous distribution function  $F$ :  $q_\alpha = F^{-1}(\alpha)$ , a negative number for small  $\alpha$

## Value at Risk

- **Definition:**  $\text{VaR}_\alpha(X) = -q_\alpha(X)$ 
  - ▶ the loss that is exceeded with probability  $\alpha$ :  $P(L > \text{VaR}_\alpha) \leq \alpha$
  - ▶ the minus sign makes VaR a positive loss; a VaR of 3% means "a loss of more than 3% happens on 1 day in 100"
- What VaR does **not** say
  - ▶ how large the loss is on the bad days: a loss of 4% and a loss of 40% beyond VaR give the same VaR
  - ▶ VaR is not the maximum loss: by construction it is exceeded on about  $\alpha$  of the days
- In money: a position of value  $W$  has  $\text{VaR} = W \times \text{VaR}(\%)/100$

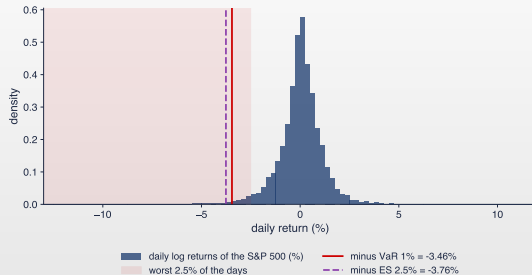
## Expected Shortfall

- **Definition:**  $ES_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$ : the average of the VaRs at all levels below  $\alpha$ 
  - ▶ for a continuous distribution:  $ES_\alpha = -E[X \mid X \leq q_\alpha(X)]$ , the average loss on the worst  $\alpha$  of the days
  - ▶ other names: CVaR (conditional VaR), TVaR (tail VaR), expected tail loss
- **Properties**
  - ▶  $ES_\alpha \geq VaR_\alpha$  at the same level  $\alpha$
  - ▶ ES looks inside the tail: two distributions with the same VaR but different tails have different ES

## Worked Example: VaR and ES from 100 Days

- A position has 100 daily returns; the five worst are  $-6.1$ ,  $-4.8$ ,  $-4.2$ ,  $-3.9$ ,  $-3.5$  (in %)
  - ▶ empirical quantile at level  $\alpha$ : the  $k$ -th smallest return,  $k = \lceil n\alpha \rceil$  ( $\lceil \cdot \rceil$ : rounding up)
- **VaR 1%**:  $k = \lceil 100 \times 0.01 \rceil = 1$ , so  $\text{VaR}_{1\%} = 6.1\%$ 
  - ▶ one observation decides VaR 1%: with 100 days the estimate is very imprecise
- **ES 2.5%**:  $k = \lceil 2.5 \rceil = 3$ ; the average of the three worst losses
  - ▶  $\text{ES}_{2.5\%} = (6.1 + 4.8 + 4.2)/3 = 5.03\%$ ;  $\text{VaR}_{2.5\%} = 4.2\%$
- A position of 1 million RON:  $\text{VaR } 1\% = 61\,000$  RON; on one day in a hundred the loss is larger

## VaR 1% and ES 2.5% of the S&P 500



- Daily log returns since 2000 (6 714 days); historical estimates: VaR 1% = 3.46%, VaR 2.5% = 2.51%, ES 2.5% = 3.76%
- Interpretation: ES 2.5% averages the whole shaded tail, including the extreme days of 2008 and 2020; VaR 1% is one point in that tail
- For a position of 1 million RON in the index: VaR 1% about 34 552 RON, ES 2.5% about 37 613 RON

## The Normal Distribution: Closed Forms

□  $X \sim N(\mu, \sigma^2)$ ,  $z_\alpha = \Phi^{-1}(\alpha)$  ( $\Phi$ ,  $\varphi$ : distribution function and density of  $N(0, 1)$ )

▶  $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ ,  $\text{ES}_\alpha = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$

▶ derivation:  $E[Z \mid Z \leq z] = -\varphi(z)/\Phi(z)$  because  $\varphi'(z) = -z\varphi(z)$

□ Numbers:  $z_{0.01} = -2.326$ ;  $z_{0.025} = -1.960$ ,  $\varphi(z_{0.025}) = 0.0584$ ,  $\varphi(z_{0.025})/0.025 = 2.338$

▶  $\text{ES } 2.5\% / \text{VaR } 1\% = 2.338/2.326 = 1.005$  (with  $\mu = 0$ ): for Normal returns the two measures are almost equal

▶ this is why ES 2.5% replaced VaR 1% in Basel: same size for Normal data, larger for heavy tails

## Worked Example: Normal VaR and ES of the S&P 500

- Sample moments of the daily returns since 2000:  $\hat{\mu} = 0.025$ ,  $\hat{\sigma} = 1.21$  (in %)
  - ▶  $\text{VaR } 1\% = -(0.025 - 2.326 \times 1.21) = -0.025 + 2.822 = 2.80\%$
  - ▶  $\text{ES } 2.5\% = -0.025 + 2.338 \times 1.21 = -0.025 + 2.836 = 2.81\%$
- Compare with the data (historical simulation):  $\text{VaR } 1\% = 3.46\%$ ,  $\text{ES } 2.5\% = 3.76\%$ 
  - ▶ the Normal VaR 1% is 19% too low: the Normal tail is too thin ([Chapter 2](#))
  - ▶ the Normal ES 2.5% is even further from the data: it ignores the extreme days
- Interpretation: two parameters cannot describe the tail; the next methods change either the distribution or the data used

## The Student-t Distribution: Closed Forms

- $X = m + s T$ ,  $T \sim t_\nu$  (location  $m$ , scale  $s$ ,  $\nu$  degrees of freedom, [Chapter 2](#));  $t_\alpha = t_\nu^{-1}(\alpha)$ ,  $f_\nu$  the density of  $t_\nu$ 
  - ▶  $\text{VaR}_\alpha = -(m + s t_\alpha)$ ,  $\text{ES}_\alpha = -m + s \frac{f_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$  (for  $\nu > 1$ )
  - ▶ the variance is  $s^2 \nu / (\nu - 2)$ : a **standardised**  $t$  (variance 1) has  $s = \sqrt{(\nu - 2)/\nu}$
- **Worked example:** S&P 500, maximum likelihood:  $\hat{m} = 0.068$ ,  $\hat{s} = 0.703$ ,  $\hat{\nu} = 2.67$ 
  - ▶  $t_{0.01} = -5.023$ :  $\text{VaR } 1\% = -(0.068 + 0.703 \times (-5.023)) = 3.46\%$
  - ▶ ES factor 5.720:  $\text{ES } 2.5\% = -0.068 + 0.703 \times 5.720 = 3.95\%$
- Interpretation: with  $\nu < 3$  the fitted tail is very heavy: VaR 1% matches the data, ES 2.5% is above the historical 3.76%

## From One Day to Ten Days

- **Square-root-of-time rule:**  $\text{VaR}^{(h)} \approx \sqrt{h} \text{VaR}^{(1)}$ 
  - ▶ exact for i.i.d. Normal returns with mean 0: the  $h$ -day return has standard deviation  $\sigma\sqrt{h}$
  - ▶ Basel 1996: 10-day VaR, usually scaled from one day with  $\sqrt{10} = 3.162$
- S&P 500, historical simulation
  - ▶ one day: 3.46%; scaled:  $3.162 \times 3.46 = 10.93\%$
  - ▶ VaR 1% of overlapping 10-day returns: 10.37%
- Why the rule fails in general
  - ▶ heavy tails: a sum of 10 heavy-tailed returns is closer to Normal (CLT, central limit theorem), so the ratio of quantiles is below  $\sqrt{10}$
  - ▶ volatility clustering: after a calm day the 10-day risk is larger than  $\sqrt{10}$  times today's, after a storm smaller ([Chapter 9](#))

## Recap: Definitions of VaR and ES

- ▣  $\text{VaR}_\alpha = -q_\alpha$ : the loss exceeded with probability  $\alpha$ ;  $\text{ES}_\alpha$ : the average loss beyond it
- ▣ Normal:  $\text{ES } 2.5\% \approx \text{VaR } 1\%$ ; heavy tails make ES 2.5% larger
- ▣ Scaling to 10 days by  $\sqrt{10}$  is a convention, not a law

## 1999: Four Axioms for a Risk Measure

- Artzner et al. (1999): a risk measure  $\rho$  maps a loss  $L$  to the capital that makes it acceptable;  $\rho$  is **coherent** if
  - ▶ **monotonicity**:  $L_1 \leq L_2$  always  $\Rightarrow \rho(L_1) \leq \rho(L_2)$
  - ▶ **translation invariance**:  $\rho(L + c) = \rho(L) + c$  for a sure amount  $c$
  - ▶ **positive homogeneity**:  $\rho(\lambda L) = \lambda \rho(L)$ ,  $\lambda > 0$
  - ▶ **subadditivity**:  $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$
- Subadditivity: diversification never adds risk; a bank can add the capital of its desks as an upper bound



Freddy Delbaen, co-author of the axioms  
(1997)

Photo: ETH-Bibliothek Zürich, Bildarchiv  
(1997); CC BY-SA 4.0; Wikimedia Commons

## Counterexample: Two Bonds, Step by Step

- Two independent bonds; each loses 100 with probability 4% (default) and 0 otherwise; level  $\alpha = 5\%$ 
  - ▶ one bond:  $P(L > 0) = 4\% \leq 5\%$ , so  $\text{VaR}_{5\%} = 0$  for each
- Both bonds:  $P(L = 200) = 0.16\%$ ,  $P(L = 100) = 7.68\%$ , so  $P(L > 0) = 7.84\% > 5\%$ 
  - ▶  $\text{VaR}_{5\%}(L_1 + L_2) = 100 > 0 + 0$ : VaR **punishes** diversification
- ES 5% (the average of the worst 5% of outcomes)
  - ▶ one bond:  $(4\% \times 100 + 1\% \times 0)/5\% = 80$
  - ▶ both:  $(0.16\% \times 200 + (5 - 0.16)\% \times 100)/5\% = 103.2 \leq 80 + 80$ : subadditive

## ES Is Coherent; VaR Only Sometimes

- **ES** satisfies all four axioms [Acerbi & Tasche \(2002\)](#)
  - ▶ reason: ES is an average of quantiles over the tail, and averaging respects sums
- **VaR** satisfies the first three; subadditivity can fail
  - ▶ fails for discrete losses (credit, the bonds above) and for extremely heavy tails (tail index below 1, [Chapter 5](#))
  - ▶ holds for elliptical distributions (Normal, Student-t) and, in practice, for most equity returns [Danielsson et al. \(2013\)](#)
- ES needs a finite mean of the losses; VaR always exists

## Elicitability: Can a Forecast Be Scored?

- A statistic is **elicitable** if some loss (scoring) function is minimised, on average, by its true value [Gneiting \(2011\)](#)
  - ▶ the mean: squared error; the median: absolute error
- **VaR is elicitable**: the quantile (pinball) loss  $S(v, x) = (\alpha - 1\{x < -v\})(x + v)$ 
  - ▶ two VaR models can be compared by their average quantile loss
- **ES alone is not elicitable** [Gneiting \(2011\)](#); the pair (VaR, ES) is [Fissler & Ziegel \(2016\)](#)
  - ▶ consequence: ES forecasts are tested together with VaR forecasts (Section 7)

## Recap: Coherent Risk Measures

- ▣ Coherence: monotonicity, translation invariance, positive homogeneity, subadditivity
- ▣ ES is coherent; VaR can penalise diversification (two bonds)
- ▣ VaR is elicitable, ES only together with VaR

## Five Unconditional Methods at a Glance

| Method                     | Assumption                                               | Weak point                                    |
|----------------------------|----------------------------------------------------------|-----------------------------------------------|
| historical simulation (HS) | the past window is a sample of tomorrow                  | few tail points; nothing beyond the worst day |
| Normal                     | returns are Normal                                       | tails far too thin                            |
| Student-t                  | returns are Student-t                                    | one tail shape for both tails                 |
| Cornish–Fisher             | Normal quantile corrected by skewness and kurtosis       | breaks down for large kurtosis                |
| EVT (POT)                  | GPD above a high threshold ( <a href="#">Chapter 5</a> ) | choice of the threshold                       |

- “Unconditional”: the same VaR every day of the window; conditional methods (Section 5) use today’s volatility
- All methods: daily log returns in % ([Chapter 1](#)), VaR and ES as positive losses

## Historical Simulation

- ▣ **Algorithm:** take the last  $n$  returns  $r_{t-n+1}, \dots, r_t$ , sort them:  $r_{(1)} \leq \dots \leq r_{(n)}$ 
  - ▶  $\widehat{\text{VaR}}_\alpha = -r_{(k)}$ ,  $k = \lceil n\alpha \rceil$ ;  $\widehat{\text{ES}}_\alpha = -\frac{1}{k} \sum_{i=1}^k r_{(i)}$
  - ▶ no distribution is assumed: the empirical distribution of the window is the forecast
- ▣ Ported from the SFE Quantlet VaRest

## Worked Example: Historical Simulation on the S&P 500

- S&P 500 since 2000:  $n = 6\,714$ 
  - ▶ VaR 1%:  $k = \lceil 67.14 \rceil = 68$ , the 68-th worst day: 3.46%
  - ▶ ES 2.5%: the average of the  $\lceil 167.85 \rceil = 168$  worst days: 3.76%
- The window  $n$  is a trade-off
  - ▶ short (250 days): reacts, but VaR 1% rests on 3 observations; long (all data): stable, but blind to today's volatility
  - ▶ **ghost effect**: a crash day changes VaR when it enters the window and again, abruptly,  $n$  days later when it leaves

## How Precise Is a Historical VaR?

- A VaR estimate is a statistic: it has a sampling error; we measure it with the bootstrap ([Chapter 2](#))
  - ▶ resample the  $n$  returns with replacement 2000 times, recompute VaR and ES, read the 2.5% and 97.5% quantiles
- S&P 500, all 6714 days: VaR 1% = 3.46%, 95% interval [3.18, 3.66]; ES 2.5% = 3.76%, [3.46, 4.07]
  - ▶ last 500 days: VaR 1% = 2.75%, [1.99, 4.96]; ES 2.5% = 2.94%, [2.14, 3.82]
- With two years of data the interval of VaR 1% is wider than 2 percentage points: VaR 1% rests on 5 observations
- Interpretation: differences between methods smaller than the bootstrap interval are not evidence that one method is better

 [SFM\\_ch10\\_var\\_methods](#)

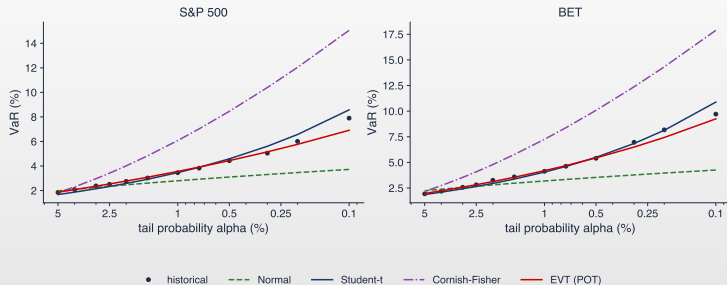
## The Cornish–Fisher Expansion

- **Cornish & Fisher (1938)**: correct the Normal quantile  $z = z_\alpha$  with the skewness  $S$  and the excess kurtosis  $K$  (kurtosis minus 3)
  - ▶  $z_{CF} = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$ ,  $\text{VaR}_\alpha = -(\mu + \sigma z_{CF})$
  - ▶ negative  $S$  and positive  $K$  push the quantile further into the left tail
- **S&P 500**:  $S = -0.35$ ,  $K = 10.7$ ,  $z_{0.01} = -2.326$ 
  - ▶  $z_{CF} = -5.03$ , so  $\text{VaR } 1\% = 6.08\%$ , against the historical 3.46%
- **Interpretation**: the expansion is a small correction around the Normal; with an excess kurtosis above 10 it overshoots badly
  - ▶ use it only when  $|S|$  and  $K$  are moderate (for example monthly returns or diversified portfolios)

## EVT: Peaks over Threshold (Chapter 5)

- Losses  $L = -r$ ; threshold  $u$  = the 90% quantile of the losses [McNeil & Frey \(2000\)](#); excesses  $L - u$  above  $u$  follow a GPD (generalised Pareto distribution) with shape  $\xi$  and scale  $\beta$ 
  - ▶  $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[ \left( \frac{n\alpha}{N_u} \right)^{-\xi} - 1 \right], \quad \text{ES}_\alpha = \frac{\text{VaR}_\alpha + \beta - \xi u}{1 - \xi} \quad (N_u: \text{ number of excesses})$
- S&P 500:  $u = 1.26\%$ ,  $N_u = 672$  (10.0% of the days),  $\hat{\xi} = 0.164$ ,  $\hat{\beta} = 0.820$ 
  - ▶ VaR 1%:  $n\alpha/N_u = 0.0999$ ,  $0.0999^{-\hat{\xi}} = 1.461$ ;  $1.26 + 4.988 \times (1.461 - 1) = 3.56\%$
  - ▶ ES 2.5% = 3.77%;  $\hat{\xi} > 0$ : a heavy (Pareto-type) tail
- EVT smooths the tail and extrapolates beyond the worst observed day: useful for VaR 0.1% and for stress tests

## VaR across Tail Probabilities



- Each line:  $\text{VaR}_\alpha$  of one method for  $\alpha$  from 5% to 0.1%; points: historical simulation
- Interpretation: at 5% all methods agree; at 1% the Normal is too low; at 0.1% Student-t and EVT stay close to the data, while Cornish-Fisher explodes

## VaR 1% by Method on Six Series

|         | HS    | Normal | Student-t | Cornish–Fisher | EVT   | $K$  | $\hat{\xi}$ |
|---------|-------|--------|-----------|----------------|-------|------|-------------|
| S&P 500 | 3.46  | 2.80   | 3.46      | 6.08           | 3.56  | 10.7 | 0.16        |
| DAX     | 4.20  | 3.26   | 3.96      | 5.42           | 4.05  | 6.1  | 0.12        |
| BET     | 4.14  | 3.19   | 4.06      | 7.25           | 4.19  | 11.1 | 0.23        |
| Bitcoin | 10.54 | 7.99   | 11.06     | 18.89          | 10.38 | 12.0 | 0.12        |
| TLV     | 5.11  | 4.01   | 4.63      | 10.19          | 4.90  | 13.5 | 0.26        |
| SNP     | 4.29  | 3.70   | 4.26      | 7.55           | 4.38  | 8.9  | 0.22        |

- Whole samples (S&P 500, DAX, BET since 2000; Bitcoin since 2014; TLV and SNP since 2010), in %;  $K$ : excess kurtosis
- HS, Student-t and EVT differ by at most 0.7 percentage points; the Normal VaR is 14–24% too low everywhere
- Cornish–Fisher gives 1.3 to 2.0 times the historical VaR: with  $K$  between 6 and 14 the expansion is outside its range

## ES 2.5% by Method on Six Series

|         | HS    | Normal | Student-t | Cornish–Fisher | EVT   | $K$  | $\hat{\xi}$ |
|---------|-------|--------|-----------|----------------|-------|------|-------------|
| S&P 500 | 3.76  | 2.81   | 3.95      | 6.65           | 3.77  | 10.7 | 0.16        |
| DAX     | 4.26  | 3.27   | 4.37      | 5.81           | 4.24  | 6.1  | 0.12        |
| BET     | 4.57  | 3.21   | 4.76      | 7.93           | 4.57  | 11.1 | 0.23        |
| Bitcoin | 10.84 | 8.03   | 13.37     | 20.69          | 10.90 | 12.0 | 0.12        |
| TLV     | 5.37  | 4.03   | 5.03      | 11.20          | 5.35  | 13.5 | 0.26        |
| SNP     | 4.70  | 3.72   | 4.57      | 8.18           | 4.71  | 8.9  | 0.22        |

- Normal: ES 2.5% equals its VaR 1%; the data: the historical ES 2.5% is 1.5–10.5% above the historical VaR 1%
- For the S&P 500, DAX, BET and Bitcoin, Student-t gives the largest ES among the sensible methods (Bitcoin: 13.37% against 10.84%): with  $\hat{\nu} < 3$  its tail is heavier than the data beyond the threshold
- Interpretation: EVT and HS agree at 2.5%; the method matters more for ES than for VaR, because ES depends on the whole tail

## Recap: Estimating VaR and ES

- HS: sorted window; Normal and Student-t: closed forms; Cornish–Fisher: moment correction; EVT: GPD tail
- On daily data: Normal too low, Cornish–Fisher too high, HS, Student-t and EVT close at 1%
- All five ignore today's volatility

## Why Condition on Today's Volatility?

- Volatility clustering (Chapters 8 and 9): a calm day is followed by calm days, a storm by storms
  - ▶ an unconditional VaR is too high in calm years and too low at the start of a crisis
- Conditional model:  $r_{t+1} = \mu + \sigma_{t+1} z_{t+1}$ ,  $z$  i.i.d. with mean 0 and variance 1
  - ▶  $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1} q_\alpha(z))$ ,  $\text{ES}_{t+1} = -\mu + \sigma_{t+1} \text{ES}_\alpha(z)$
  - ▶ two ingredients: the volatility forecast  $\sigma_{t+1}$  and the distribution of  $z$
- Two choices for the distribution of  $z$ 
  - ▶ **GARCH-t**: standardised Student-t with  $\nu$  estimated with the GARCH(1,1) [Bollerslev \(1986\)](#)
  - ▶ **FHS** (filtered historical simulation): the empirical distribution of the standardised residuals  $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$  [Hull & White \(1998\)](#); [Barone-Adesi, Giannopoulos & Vosper \(1999\)](#)

## Worked Example: the S&P 500 on 18 September 2026

- GARCH(1,1)-t on all data:  $\hat{\mu} = 0.079$ ,  $\hat{\alpha} = 0.123$ ,  $\hat{\beta} = 0.872$ ,  $\hat{\nu} = 6.26$ ; forecast for the next day  $\sigma_{t+1} = 0.714\%$
- **GARCH-t**:  $q_{0.01}(z) = t_{\nu}^{-1}(0.01)\sqrt{(\nu - 2)/\nu} = -2.557$ ;  $ES_{2.5\%}(z) = 2.644$ 
  - ▶ VaR 1% =  $-(0.079 + 0.714 \times (-2.557)) = 1.75\%$ ;  $ES\ 2.5\% = -0.079 + 0.714 \times 2.644 = 1.81\%$
- **FHS**: the 1% quantile of the 6714 standardised residuals is  $-2.787$ ; their ES 2.5% is 2.952
  - ▶ VaR 1% =  $-(0.079 + 0.714 \times (-2.787)) = 1.91\%$ ;  $ES\ 2.5\% = 2.03\%$
- Interpretation: a calm market: the conditional VaR 1% is about half of the unconditional 3.46%; the standardised residuals have a heavier left tail than the fitted t

## Next-Day VaR and ES on Six Series

|         | HS (all) | VaR 1%<br>GARCH-t | FHS  | HS (all) | ES 2.5%<br>GARCH-t | FHS  | $\sigma_{t+1}$ |
|---------|----------|-------------------|------|----------|--------------------|------|----------------|
| S&P 500 | 3.46     | 1.75              | 1.91 | 3.76     | 1.81               | 2.03 | 0.714          |
| DAX     | 4.20     | 2.31              | 2.48 | 4.26     | 2.38               | 2.55 | 0.945          |
| BET     | 4.14     | 4.51              | 4.81 | 4.57     | 4.72               | 5.04 | 1.763          |
| Bitcoin | 10.54    | 7.37              | 7.99 | 10.84    | 8.11               | 8.57 | 2.829          |
| TLV     | 5.11     | 7.18              | 7.33 | 5.37     | 7.53               | 8.00 | 2.791          |
| SNP     | 4.29     | 5.06              | 5.39 | 4.70     | 5.31               | 5.65 | 1.966          |

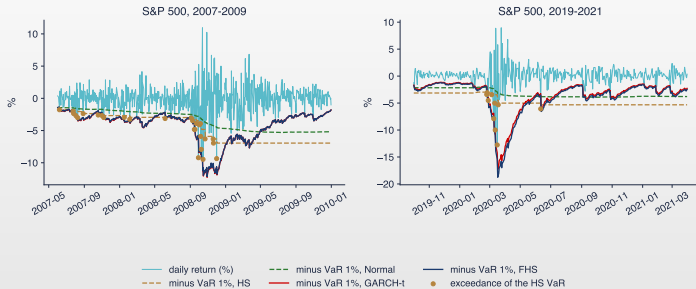
- Forecasts for the day after 18 September 2026, in %; HS (all): the unconditional historical values of the previous section
- S&P 500, DAX, Bitcoin: calm, the conditional values are below the unconditional ones
- Interpretation: BET, TLV and SNP: the current volatility is high, so the conditional VaR is above the long-run historical VaR; the same method gives opposite messages on different markets

## Rolling Forecasts: the Design

- ▣ **Out of sample:** the forecast for day  $t$  uses only data up to day  $t - 1$ 
  - ▶ S&P 500, DAX, BET: every day since January 2005 (2008 included); Bitcoin: since January 2017
- ▣ Four methods
  - ▶ **HS** and **Normal**: the last 500 days (about two years)
  - ▶ **GARCH-t** and **FHS**: GARCH(1,1)-t re-estimated every 250 days on all past data;  $\sigma_t$  updated every day
- ▣ Each day: VaR 1% (for the backtests), VaR 2.5% and ES 2.5% (for the ES backtest)
- ▣ Ported from SFEVaRtimeplot and SFEVaRbank

 SFM\_ch10\_conditional\_var

## VaR 1% through 2008 and 2020



- S&P 500: daily returns and minus the VaR 1% forecasts of the four methods; dots: exceptions of the HS VaR
- HS and Normal react months late: the 500-day window changes slowly; GARCH-t and FHS jump within days
- Interpretation: after the crisis HS stays high for two years (the ghost effect), while GARCH-t returns to normal levels

## Stress Periods: Exceptions in 2008 and 2020

|         | Sep 2008 – Mar 2009 |        |         |     |      | Feb – May 2020 |        |         |     |      |
|---------|---------------------|--------|---------|-----|------|----------------|--------|---------|-----|------|
|         | HS                  | Normal | GARCH-t | FHS | exp. | HS             | Normal | GARCH-t | FHS | exp. |
| S&P 500 | 15                  | 21     | 5       | 5   | 1.5  | 9              | 12     | 3       | 3   | 0.7  |
| DAX     | 6                   | 18     | 3       | 3   | 1.5  | 8              | 13     | 3       | 3   | 0.7  |
| BET     | 8                   | 14     | 4       | 4   | 1.4  | 5              | 10     | 4       | 4   | 0.7  |

- Exceptions of the VaR 1%: days with  $r_t < -\text{VaR}_t$ ; exp.: 1% of the days in the period
- In 2008 the S&P 500 Normal VaR was exceeded on 21 of 146 days, HS on 15; GARCH-t and FHS on 5
- Interpretation: no method is perfect in a crisis, but the conditional ones fail far less: the volatility forecast matters more than the tail shape

## Which Method When?

| Situation                                                             | Method                      | Reason                                                         |
|-----------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------|
| daily VaR of a liquid position                                        | GARCH-t, FHS                | follow the volatility; FHS keeps the empirical tail            |
| long history, quick check<br>very small $\alpha$ (0.1%), stress tests | HS<br>EVT                   | no model, easy to explain<br>extrapolates beyond the worst day |
| portfolio of many assets                                              | variance–covariance, copula | dependence; the t copula for joint crashes                     |
| moderate skewness and kurtosis                                        | Cornish–Fisher              | a quick correction of the Normal                               |

- Never only the Normal for daily data; always report the method, the window and the level
- Whatever the method: backtest it (Section 7)

## Recap: Conditional VaR

- $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1}q_\alpha(z))$ : GARCH-t (parametric  $z$ ) or FHS (empirical  $z$ )
- Conditional VaR follows the storms; window-based VaR lags and keeps ghosts
- In 2008 and 2020 the conditional methods had a few times fewer exceptions

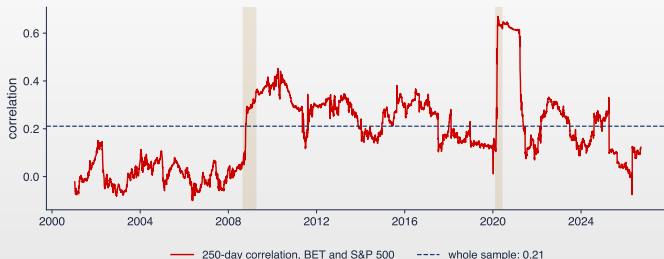
## A BET and S&P 500 Portfolio

- Weights  $w = (0.5, 0.5)$ ; simple returns  $R_t$  in %, so that  $R_{p,t} = w_1 R_{1,t} + w_2 R_{2,t}$  exactly
  - ▶ prices joined on the 6 466 common trading days since 2000, then returns
  - ▶ each index in its own currency: the currency risk of a RON investor is left out
- Daily data: BET  $\hat{\mu} = 0.076$ ,  $\hat{\sigma} = 1.42$ ; S&P 500  $\hat{\mu} = 0.034$ ,  $\hat{\sigma} = 1.23$ ; correlation  $\hat{\rho} = 0.21$ 
  - ▶ the BVB closes before New York opens: part of the joint reaction appears one day later, so same-day correlation understates the link
- Question: is the portfolio VaR smaller than the average of the two VaRs, and by how much?

## The Variance–Covariance Method

- Portfolio:  $\mu_p = w^\top \mu$ ,  $\sigma_p^2 = w^\top \Sigma w = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$  ( $\Sigma$ : covariance matrix)
  - ▶ if returns are jointly Normal:  $\text{VaR}_p = -(\mu_p + z_\alpha \sigma_p)$  (the RiskMetrics method J.P. Morgan/Reuters (1996))
- **Step by step:**  $\sigma_p^2 = 0.712^2 + 0.614^2 + 2 \times 0.5 \times 0.5 \times 0.369 = 1.068$ ,  $\sigma_p = 1.033$ 
  - ▶  $\text{VaR } 1\% = -(0.055 - 2.326 \times 1.033) = 2.35\%$ ; the weighted sum of the two Normal VaRs:  
 $0.5 \times 3.23 + 0.5 \times 2.82 = 3.03\%$
  - ▶ **diversification benefit:** 22% of the risk disappears; with  $\rho = 1$  it would be zero
- Euler contributions  $w_i(\Sigma w)_i / \sigma_p$  add up to  $\sigma_p$ : BET 56%, S&P 500 44% of the portfolio risk

## Correlation Rises in Crises



- 250-day correlation of BET and S&P 500 daily returns; shaded: the 2008 and 2020 stress periods
- Calm 2017: 0.20; September 2008 – March 2009: 0.37; February – May 2020: 0.71; maximum 0.67 on 25 March 2020
- Interpretation: diversification is weakest exactly when it is needed: a VaR built on the average correlation is too optimistic in a crisis

## Correlation Is Not Tail Dependence

- Pearson correlation measures linear co-movement over all days; it is dominated by ordinary days Embrechts, McNeil & Straumann (2002)
  - ▶ rank correlations (Spearman, Kendall's  $\tau$ ) use only the order of the returns: Kendall  $\hat{\tau} = 0.100$ , Spearman 0.15
- **Lower tail dependence:**  $\lambda_L = \lim_{q \rightarrow 0} P(U_1 \leq q \mid U_2 \leq q)$ ,  $U_i = F_i(X_i)$  (the returns on a uniform scale)
  - ▶ the chance that BET has one of its worst days given that the S&P 500 has one of its worst days
  - ▶ data:  $q = 5\%$ : 0.22;  $q = 1\%$ : 0.19; under independence both would equal  $q$
- Joint crash (both below their own 1% quantile): 0.19% of the days, against 0.01% under independence

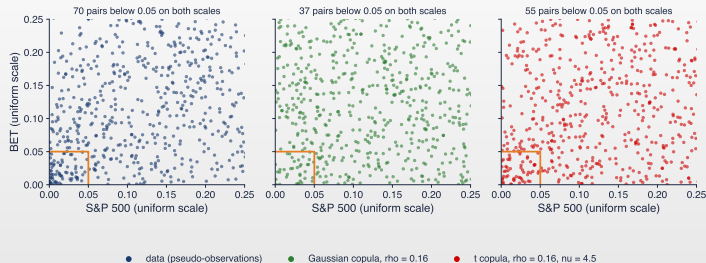
## Copulas: a Light Introduction

- **Sklar's theorem:** every joint distribution function can be written as  $F(x_1, x_2) = C(F_1(x_1), F_2(x_2))$  [McNeil, Frey & Embrechts \(2015\)](#)
  - ▶  $F_1, F_2$ : the marginal distributions (each asset alone);  $C$ : the **copula**, a joint distribution of uniforms on  $[0, 1]^2$
  - ▶ the copula holds all the dependence, the margins all the individual tails
- Practical recipe
  - ▶ margins: empirical or fitted (Student-t, EVT); dependence: a copula fitted to the **pseudo-observations**  
 $\hat{U}_{i,t} = \text{rank}(X_{i,t}) / (n + 1)$
  - ▶ simulate  $(U_1, U_2)$  from the copula, map back with  $F_i^{-1}$ , compute the portfolio return, read VaR and ES
- Textbook: [Franke, Härdle & Hafner \(2019\)](#), Ch. 17

## Gaussian and t Copulas

- **Gaussian copula:** the dependence of a bivariate Normal with correlation  $\rho$ ;  $\lambda_L = 0$  for  $\rho < 1$ 
  - ▶ extreme days become independent in the limit: joint crashes are rare
- **t copula:** the dependence of a bivariate Student-t with  $\rho$  and  $\nu$  Demarta & McNeil (2005)
  - ▶  $\lambda_L = 2 t_{\nu+1} \left( - \sqrt{(\nu+1)(1-\rho)/(1+\rho)} \right) > 0$ : crashes cluster across assets
- Fit on BET and S&P 500:  $\rho = \sin(\pi \hat{\tau}/2) = 0.157$  (valid for both copulas);  $\hat{\nu} = 4.50$  by maximum likelihood
  - ▶ log-likelihood: Gaussian 92.1, t 219.5; LR = 254.9, far above  $\chi^2_{0.95}(1) = 3.84$ : the t copula wins
  - ▶ implied tail dependence  $\lambda_L = 0.096$ , against 0 for the Gaussian copula

## Joint Bad Days: Data and Two Copulas



- Lower-left corner of the uniform scale: pseudo-observations and samples of the same size (6 466) from the fitted copulas
- Days on which both indices are among their worst 5%: data 70, Gaussian 37, t 55; independence: about 16
- Interpretation: the Gaussian copula halves the number of joint bad days; the t copula is much closer to the data

## Copula VaR: the Simulation Step by Step

- ▣ 1. Draw  $(Z_1, Z_2)$  from a bivariate Normal with correlation  $\rho$ 
  - ▶ Gaussian copula:  $U_i = \Phi(Z_i)$
  - ▶ t copula: draw  $W \sim \chi_\nu^2$ , set  $U_i = t_\nu(Z_i \sqrt{\nu/W})$ : the common factor  $\sqrt{\nu/W}$  makes extremes happen together
- ▣ 2. Map to returns with the empirical quantile functions:  $R_i = \hat{F}_i^{-1}(U_i)$
- ▣ 3. Portfolio return  $R_p = 0.5R_1 + 0.5R_2$ ; repeat 200 000 times
- ▣ 4. VaR 1% and ES 2.5% by historical simulation on the simulated  $R_p$
- ▣ Same margins in both copulas: any difference in VaR comes from the dependence alone

## Portfolio VaR 1% and ES 2.5% by Method

| Method                             | VaR 1% | ES 2.5% | P(joint crash at 1%) |
|------------------------------------|--------|---------|----------------------|
| variance–covariance (Normal)       | 2.35   | 2.36    | –                    |
| Gaussian copula, empirical margins | 2.75   | 2.93    | 0.03%                |
| t copula, empirical margins        | 2.81   | 3.03    | 0.13%                |
| historical simulation (the data)   | 3.08   | 3.30    | 0.19%                |

- In %, 50/50 portfolio, 6 466 common days; copulas: 200 000 simulated days
- The Normal method misses both the heavy margins and the tail dependence; the copulas fix the margins, the t copula also the joint crashes
- Interpretation: the t copula still understates the historical VaR: the BET–S&P 500 link is stronger in crises than any constant copula can describe

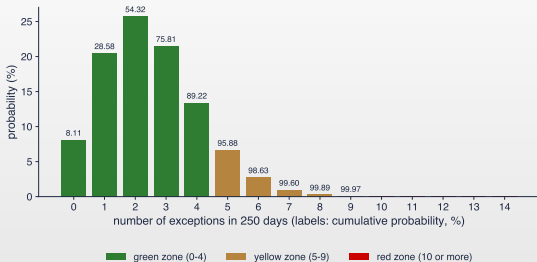
## Recap: Portfolio VaR and Dependence

- ▣ Variance–covariance:  $\sigma_p^2 = w^\top \Sigma w$ ; diversification lowers VaR when  $\rho < 1$
- ▣ Correlation rises in crises; tail dependence is what matters for joint crashes
- ▣ Copula = dependence on the uniform scale; the t copula has tail dependence, the Gaussian has none

## The Idea: the Hit Sequence

- **Exception** (hit): a day with  $r_t < -\text{VaR}_t$ ;  $I_t = 1\{r_t < -\text{VaR}_t\}$ 
  - ▶ the VaR must be the forecast made the day before: no look-ahead
- A correct VaR at level  $\alpha$ : the  $I_t$  are i.i.d. Bernoulli( $\alpha$ ) [Christoffersen \(1998\)](#)
  - ▶ **unconditional coverage**:  $P(I_t = 1) = \alpha$ : the right number of exceptions
  - ▶ **independence**: an exception today does not make one tomorrow more likely
- Number of exceptions in  $n$  days:  $x = \sum_t I_t \sim \text{Binomial}(n, \alpha)$ , mean  $n\alpha$ , variance  $n\alpha(1 - \alpha)$

## Exceptions in 250 Days under a Correct VaR 1%



- Binomial(250, 0.01): expected 2.5 exceptions; 0 exceptions with probability 8.11%, 5 or more with probability  $100 - 89.22 = \text{about } 11\%$
- Interpretation: one year of data separates a good model from a bad one only roughly: a correct model lands in the yellow zone about one year in nine

## Kupiec's POF Test

- **POF** (proportion of failures) **Kupiec (1995)**:  $H_0: P(I_t = 1) = \alpha$ ;  $\hat{\pi} = x/n$ 
  - ▶  $LR_{uc} = -2[(n-x)\ln(1-\alpha) + x\ln\alpha - (n-x)\ln(1-\hat{\pi}) - x\ln\hat{\pi}] \sim \chi^2(1)$  under  $H_0$
  - ▶ a likelihood-ratio test (**Chapter 6**); reject at 5% if  $LR_{uc} > 3.84$
- **Step by step**:  $n = 250$ ,  $x = 7$ ,  $\hat{\pi} = 0.028$ 
  - ▶  $\ln L_0 = 243 \ln 0.99 + 7 \ln 0.01 = -34.68$ ;  $\ln L_1 = 243 \ln 0.972 + 7 \ln 0.028 = -31.93$
  - ▶  $LR_{uc} = -2(-34.68 - (-31.93)) = 5.50$ ,  $p = 0.019$ : rejected at 5%
- **Low power with one year of data**
  - ▶ in 250 days, any count from 1 to 6 is accepted at 5%: a model with a true rate of 2% passes with probability 76%
  - ▶ too few exceptions are also rejected: an overly prudent VaR wastes capital

## Kupiec on 21 Years of S&P 500 Forecasts

- GARCH-t VaR 1%, 3 January 2005 – 18 September 2026:  $n = 5\,460$  days,  $x = 95$  exceptions,  $\hat{\pi} = 0.0174$ 
  - ▶ expected  $n\alpha = 55$ ; a 95% binomial range for the rate under  $H_0$ : 0.75–1.26%
- **Step by step**
  - ▶  $\ln L_0 = 5365 \ln 0.99 + 95 \ln 0.01 = -491.4$
  - ▶  $\ln L_1 = 5365 \ln(1 - 0.0174) + 95 \ln 0.0174 = -479.0$
  - ▶  $LR_{uc} = -2(\ln L_0 - \ln L_1) = 24.73$ ,  $p < 0.001$
- Interpretation: with 21 years of data even a rate of 1.74% instead of 1% is clearly rejected; the missing leverage effect (Chapter 9) is a likely cause

## Christoffersen's Independence Test

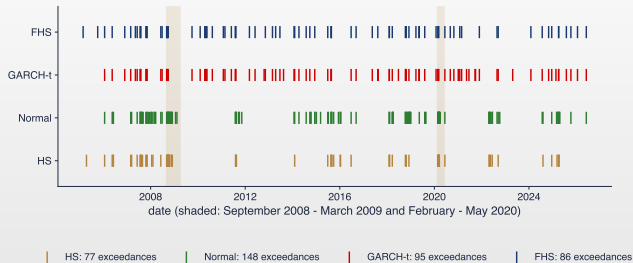
- Count the transitions:  $n_{ij}$  = number of days with  $I_{t-1} = i$  and  $I_t = j$ ;  $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$ ,  $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$ 
  - ▶  $H_0$ :  $\pi_{01} = \pi_{11}$  (an exception today does not change the chance of one tomorrow) Christoffersen (1998)
  - ▶  $LR_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \sim \chi^2(1)$
- **Conditional coverage:**  $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$ ; reject at 5% if  $LR_{cc} > 5.99$
- **Example:** S&P 500, HS VaR 1%, 5 460 days:  $n_{00} = 5313$ ,  $n_{01} = 69$ ,  $n_{10} = 69$ ,  $n_{11} = 8$ 
  - ▶  $\hat{\pi}_{01} = 1.28\%$ ,  $\hat{\pi}_{11} = 10.4\%$ : after an exception the next one is 8 times more likely
  - ▶  $LR_{ind} = 19.43$  ( $p < 0.001$ ): the exceptions cluster

## The Basel Traffic Light

| x    | Zone   | Cum. (%) | Plus | Factor |
|------|--------|----------|------|--------|
| 0    | green  | 8.11     | 0.00 | 3.00   |
| 1    | green  | 28.58    | 0.00 | 3.00   |
| 2    | green  | 54.32    | 0.00 | 3.00   |
| 3    | green  | 75.81    | 0.00 | 3.00   |
| 4    | green  | 89.22    | 0.00 | 3.00   |
| 5    | yellow | 95.88    | 0.40 | 3.40   |
| 6    | yellow | 98.63    | 0.50 | 3.50   |
| 7    | yellow | 99.60    | 0.65 | 3.65   |
| 8    | yellow | 99.89    | 0.75 | 3.75   |
| 9    | yellow | 99.97    | 0.85 | 3.85   |
| ≥ 10 | red    | 99.99    | 1.00 | 4.00   |

- **Basel Committee (1996b)**: count the exceptions of the one-day VaR 1% over the last 250 days
  - ▶ green: no action; yellow: the multiplier grows by the plus factor; red: the model is presumed wrong
- Capital =  $(3 + \text{plus}) \times$  the 60-day average of the 10-day VaR 1% (or yesterday's value, if larger)
  - ▶ zones from the binomial: yellow starts where the cumulative probability first exceeds 95%, red where it reaches 99.99%
- x: exceptions in 250 days; Cumul.:  $P(X \leq x)$  under Binomial(250, 0.01)
- FRTB keeps the count of VaR 1% exceptions at bank and desk level **Basel Framework, MAR32**

## When Do the Exceptions Happen?



- S&P 500, 2005–2026: each mark is a day on which the VaR 1% of the method was exceeded; expected about 55 per method
- Interpretation: HS and Normal exceptions come in bursts (2007–2008, 2011, 2022); GARCH-t and FHS exceptions are spread over time, but there are more of them in calm years

## Backtest Results: S&P 500 and DAX

|         | Method  | x   | rate (%) | p POF   | p ind.  | p c.c.  | max 250 d. | Z <sub>2</sub> (p) |
|---------|---------|-----|----------|---------|---------|---------|------------|--------------------|
| S&P 500 | HS      | 77  | 1.41     | 0.004   | < 0.001 | < 0.001 | 18         | -0.39 (< 0.001)    |
|         | Normal  | 148 | 2.71     | < 0.001 | < 0.001 | < 0.001 | 29         | -1.15 (< 0.001)    |
|         | GARCH-t | 95  | 1.74     | < 0.001 | 0.114   | < 0.001 | 10         | -0.54 (< 0.001)    |
|         | FHS     | 86  | 1.58     | < 0.001 | 0.060   | < 0.001 | 10         | -0.20 (0.013)      |
| DAX     | HS      | 67  | 1.21     | 0.121   | < 0.001 | < 0.001 | 12         | -0.26 (0.002)      |
|         | Normal  | 133 | 2.41     | < 0.001 | < 0.001 | < 0.001 | 19         | -0.87 (< 0.001)    |
|         | GARCH-t | 84  | 1.52     | < 0.001 | 0.549   | 0.001   | 9          | -0.49 (< 0.001)    |
|         | FHS     | 77  | 1.40     | 0.005   | 0.418   | 0.015   | 9          | -0.18 (0.014)      |

- VaR 1% since 3 January 2005 (about 5 500 days); x: exceptions; p ind.: Christoffersen independence; p c.c.: conditional coverage; max 250 d.: the worst 250-day count
- Normal: rejected everywhere; HS: exceptions cluster; GARCH-t and FHS: independent exceptions, but a rate of 1.4–1.7%

## Backtest Results: BET and Bitcoin

|         | Method  | x   | rate (%) | p POF   | p ind.  | p c.c.  | max 250 d. | $Z_2$ (p)       |
|---------|---------|-----|----------|---------|---------|---------|------------|-----------------|
| BET     | HS      | 66  | 1.22     | 0.123   | 0.001   | 0.002   | 14         | -0.27 (0.001)   |
|         | Normal  | 122 | 2.25     | < 0.001 | < 0.001 | < 0.001 | 21         | -0.83 (< 0.001) |
|         | GARCH-t | 65  | 1.20     | 0.158   | 0.807   | 0.358   | 9          | -0.22 (0.007)   |
|         | FHS     | 65  | 1.20     | 0.158   | 0.240   | 0.185   | 7          | -0.18 (0.021)   |
| Bitcoin | HS      | 33  | 0.93     | 0.672   | 0.316   | 0.553   | 6          | -0.07 (0.224)   |
|         | Normal  | 69  | 1.94     | < 0.001 | 0.057   | < 0.001 | 16         | -0.56 (< 0.001) |
|         | GARCH-t | 48  | 1.35     | 0.045   | 0.683   | 0.123   | 8          | -0.41 (< 0.001) |
|         | FHS     | 24  | 0.68     | 0.040   | 0.155   | 0.044   | 6          | 0.12 (0.834)    |

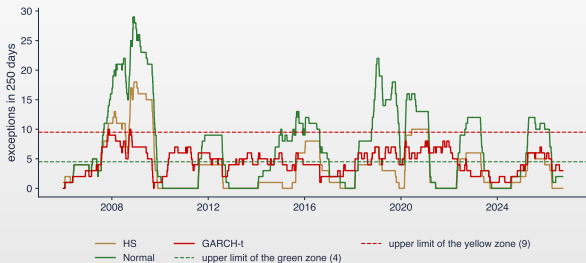
- VaR 1% since 3 January 2005 (Bitcoin since 1 January 2017); x: exceptions; p ind.: Christoffersen independence; p c.c.: conditional coverage; max 250 d.: the worst 250-day count
- BET: GARCH-t and FHS pass all three tests; Bitcoin: HS passes, FHS is too prudent (0.68%)

## Ranking the Forecasts, and Today's Zone

|         | average quantile loss $\times 100$ |        |         |       | exceptions, last 250 days |        |         |     |
|---------|------------------------------------|--------|---------|-------|---------------------------|--------|---------|-----|
|         | HS                                 | Normal | GARCH-t | FHS   | HS                        | Normal | GARCH-t | FHS |
| S&P 500 | 5.17                               | 5.80   | 3.71    | 3.68  | 0                         | 2      | 3       | 3   |
| DAX     | 5.15                               | 5.52   | 4.08    | 4.05  | 1                         | 3      | 3       | 3   |
| BET     | 6.04                               | 6.26   | 4.59    | 4.62  | 4                         | 6      | 3       | 2   |
| Bitcoin | 13.57                              | 14.52  | 13.53   | 13.75 | 3                         | 4      | 2       | 2   |

- Tests say whether a model is acceptable; a **scoring function** says which of two models is better
  - VaR is elicitable: the average quantile loss  $\frac{1}{n} \sum_t (\alpha - l_t)(r_t + \text{VaR}_t)$  ranks the forecasts [Gneiting \(2011\)](#)
- The equity indices: GARCH-t and FHS have the smallest loss, the Normal the largest; Bitcoin: the four methods are close
- Interpretation: on the last 250 days every model is in the green zone except the Normal VaR of the BET (6, yellow): a recent window says little about a model

## The Traffic Light through Time



- S&P 500: number of VaR 1% exceptions in the last 250 days, for every day since 2006; dashed: the zone limits
- Share of days in the red zone: Normal 33%, HS 12%, GARCH-t 0%; worst count: 29, 18 and 10
- Interpretation: a bank using the Normal or HS VaR would have paid the red-zone penalty for most of 2008–2009; GARCH-t almost never reaches the red zone

## Backtesting ES: Acerbi and Székely

- ES is not elicitable alone, but it can be tested together with VaR [Acerbi & Székely \(2014\)](#)

- ▶  $Z_2 = \frac{1}{T\alpha} \sum_{t=1}^T \frac{r_t l_t}{ES_t} + 1, \quad l_t = 1\{r_t < -\text{VaR}_t\}, \alpha = 2.5\%$

- ▶ if VaR and ES are right,  $E[Z_2] = 0$ ;  $Z_2 < 0$ : the losses beyond VaR are larger than the forecast ES, or too frequent

- p-value by simulation: draw returns from each day's forecast distribution, recompute  $Z_2$  1000 times

- ▶ S&P 500: Normal  $Z_2 = -1.15$ , HS  $-0.39$ , GARCH-t  $-0.54$ , FHS  $-0.20$  (p 0.013)

- Interpretation: all four ES 2.5% forecasts are too small for the equity indices; FHS is closest, because it takes the tail of  $z$  from the data

## Case Study: VaR Models at Commercial Banks

- Berkowitz & O'Brien (2002): the first study of the VaR forecasts that large US banks actually reported, with their daily trading P&L
  - ▶ design: the banks' VaR 1% against the realised P&L; a simple GARCH model of each bank's P&L as the benchmark
- Findings
  - ▶ the banks' VaR was conservative on average, yet the exceptions that did occur came in clusters
  - ▶ the GARCH benchmark gave lower VaRs with comparable coverage: it followed changes in volatility better
- The same pattern as in our data: window-based VaR is either too high in calm times or too slow in storms; conditional VaR adapts
- Further reading on forecasting the quantile directly: CAViaR Engle & Manganelli (2004)

## Recap: Backtesting

- ▣ Exceptions should be Bernoulli( $\alpha$ ) and independent: Kupiec tests the rate, Christoffersen the clustering
- ▣ Basel traffic light: 0–4 green, 5–9 yellow, 10 or more red in 250 days
- ▣ ES is tested with VaR ( $Z_2$ ); conditional methods pass more often than window methods

## An Open Question

- **Does the tail dependence between the BVB and world markets jump in crises, and does it come back?**
  - ▶ today: correlation 0.20 in a calm year, 0.71 in spring 2020; a constant t copula gives  $\lambda_L = 0.096$
  - ▶ a portfolio VaR with a constant copula is still below the historical one (previous section)
- Why it is open: crises are few, the BVB trades at other hours, and tail estimates need many observations
- Related work on entropy as a risk measure, as further reading: [Pele, Lazar & Dufour \(2017\)](#); [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- AI tools can speed up such a study; they do not replace checking it [Wang et al. \(2023\)](#)

## How AI Could Help

- ▣ **Literature:** a list of studies on time-varying copulas and on contagion between emerging and developed markets
- ▣ **Code:** a first draft of rolling t-copula estimates (250 days) for BET and S&P 500, with  $\rho$ ,  $\nu$  and  $\lambda_L$  over time
- ▣ **Robustness:** two-day returns (different trading hours), DAX instead of S&P 500, other windows
- ▣ Example prompt
  - Write Python code that fits a bivariate t copula by maximum likelihood to the pseudo-observations of daily BET and S&P 500 returns on rolling windows of 250 days moved by 21 days, and plots rho, nu and the lower tail dependence over time.

## What to Check

- ▣ The VaR convention: “VaR 1%” is the loss exceeded with probability 1%, a positive number; an answer that names the complementary probability or reports a negative VaR mixes conventions
- ▣ Prices joined on common days before computing returns; the BVB and New York trade at different hours
- ▣ Pseudo-observations: ranks divided by  $n + 1$ , not by  $n$  (otherwise the copula density is infinite at 1)
- ▣ Tail dependence from 250 days rests on a handful of joint bad days: report its uncertainty (bootstrap, [Chapter 2](#))
- ▣ Overlapping windows: consecutive estimates are not independent evidence
- ▣ References: every cited paper must exist; check the DOI

## Project Idea

- ▣ **Question:** does a time-varying copula give a better portfolio VaR for a BVB and world-equity portfolio than a constant one?
  - ▶ data: BET, BET-TR, S&P 500, DAX since 2005 (EODHD)
- ▣ Steps
  - ▶ GARCH-t margins for each index ([Chapter 9](#)); constant and rolling t copula for the dependence
  - ▶ one-day VaR 1% and ES 2.5% of a 50/50 portfolio, out of sample since 2008
  - ▶ Kupiec, Christoffersen and  $Z_2$  for both versions, with special attention to 2008, 2020 and 2022
- ▣ Deliverable: one table, one chart, and a paragraph on what the data can and cannot show
- ▣ Declare any AI use, and list the errors of the AI that you corrected

## Key Takeaways

- ▣  $\text{VaR } 1\% = -q_{1\%}$ : the loss exceeded on one day in a hundred;  $\text{ES } 2.5\%$ : the average loss on the worst 2.5% of days
- ▣ ES is coherent; VaR can penalise diversification; VaR is elicitable, ES only together with VaR
- ▣ The Normal VaR is 14–24% too low on daily data; HS, Student-t and EVT agree at 1%; Cornish–Fisher fails for large kurtosis
- ▣ Conditional VaR (GARCH-t, FHS) follows the storms and passes backtests that window methods fail
- ▣ Portfolio risk: correlation rises in crises; tail dependence (t copula) matters for joint crashes
- ▣ Backtest with Kupiec (rate), Christoffersen (clustering), the traffic light (regulation) and  $Z_2$  (ES)

## Key Formulas

| Quantity       | Formula                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------|
| VaR            | $\text{VaR}_\alpha = -q_\alpha(X)$ ; HS: $-r(\lceil n\alpha \rceil)$                                  |
| ES             | $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$ ; HS: $-\frac{1}{k} \sum_{i \leq k} r(i)$ |
| Normal         | $-(\mu + \sigma z_\alpha)$ , $-\mu + \sigma \varphi(z_\alpha)/\alpha$                                 |
| EVT (POT)      | $u + \frac{\beta}{\xi} [(n\alpha/N_u)^{-\xi} - 1]$ , ES = $(\text{VaR} + \beta - \xi u)/(1 - \xi)$    |
| Conditional    | $-(\mu + \sigma_{t+1} q_\alpha(z))$                                                                   |
| Portfolio      | $\sigma_p^2 = w^\top \Sigma w$                                                                        |
| t copula       | $\lambda_L = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)})$                                            |
| Kupiec         | $LR_{uc} = -2 \ln[(1-\alpha)^{n-x} \alpha^x / ((1-\hat{\pi})^{n-x} \hat{\pi}^x)] \sim \chi^2(1)$      |
| Christoffersen | $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$                                                         |
| Acerbi-Székely | $Z_2 = \frac{1}{T\alpha} \sum_t r_t l_t / \text{ES}_t + 1$                                            |

## Check Yourself

- **Question:** a desk reports “VaR 1% =  $-2.4\%$ ”. What is wrong?
  - ▶ **Answer:** the sign:  $-2.4\%$  is the 1% quantile  $q_{0.01}$ ;  $\text{VaR } 1\% = -q_{0.01} = 2.4\%$ , a loss
- **Question:** a model has 3 exceptions in 250 days, all in the same week. Which test detects the problem?
  - ▶ **Answer:** not Kupiec (3 is close to 2.5), but Christoffersen's independence test
- **Question:** two portfolios have the same VaR 1%. Can their ES 2.5% differ?
  - ▶ **Answer:** yes: ES depends on the whole tail beyond the quantile, VaR only on one point
- Next: [Chapter 11](#), the fractal markets hypothesis and long memory

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